

Macroeconomic Health: GDP, Growth, and Inequality

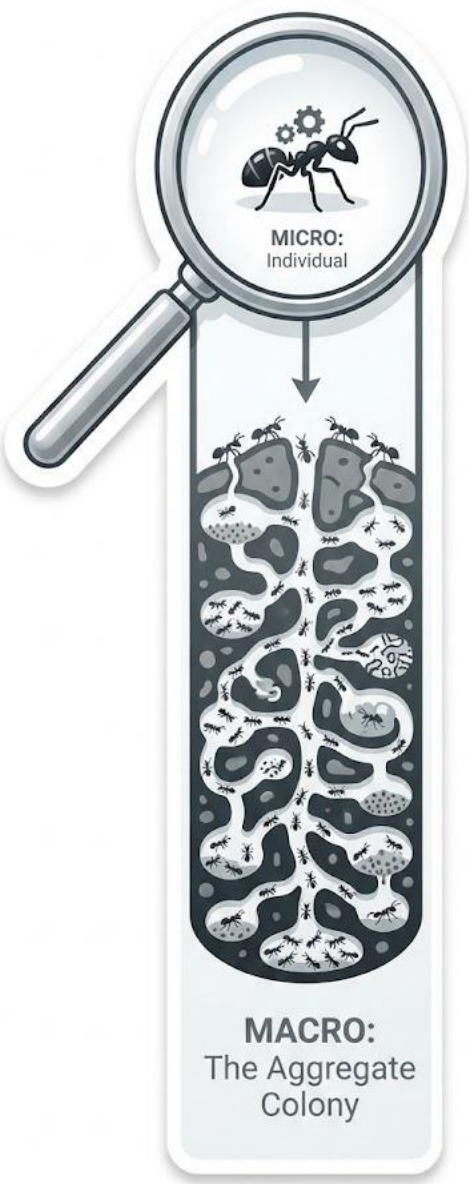
Gross Domestic Product (GDP), Economic Growth, Income Distribution, Local Economics

ITGE275 Economics

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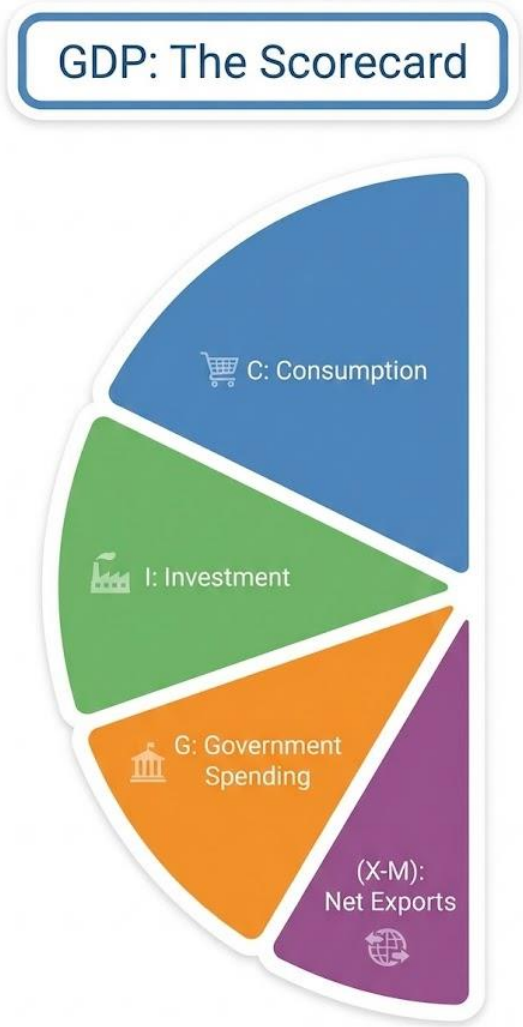
Micro vs. Macro

- **Microeconomics:** Studies the decision-making of individuals and firms (e.g., Pricing a cup of coffee, optimizing a factory).
- **Macroeconomics:** Studies the behavior of the economy as a whole (e.g., Inflation, Unemployment, Economic Growth).
- **The Aggregate:** Macro is the sum of millions of micro-decisions, but it has its own emergent properties (like Recessions).



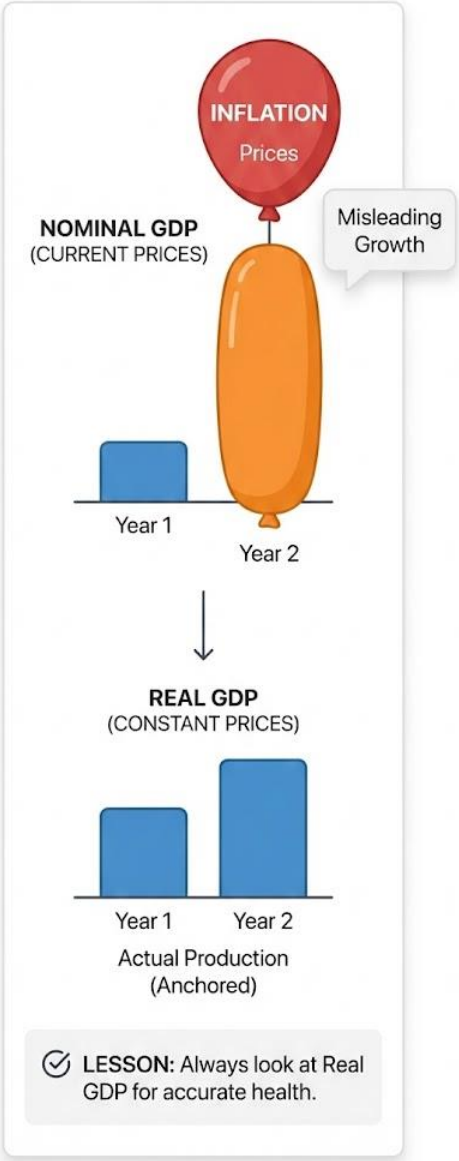
The Scorecard: Gross Domestic Product (GDP)

- **Definition:** The total market value of all final goods and services produced within a country in a given period.
- **Why it matters:** It is the primary metric for the "Size" and "Health" of an economy.
- **The Formula:** ***GDP = C + I + G + (X - M)***
  - **C:** Consumption (Spending by households).
  - **I:** Investment (Spending by businesses on capital).
  - **G:** Government Spending (Roads, Military).
  - **(X-M):** Net Exports (Exports - Imports).



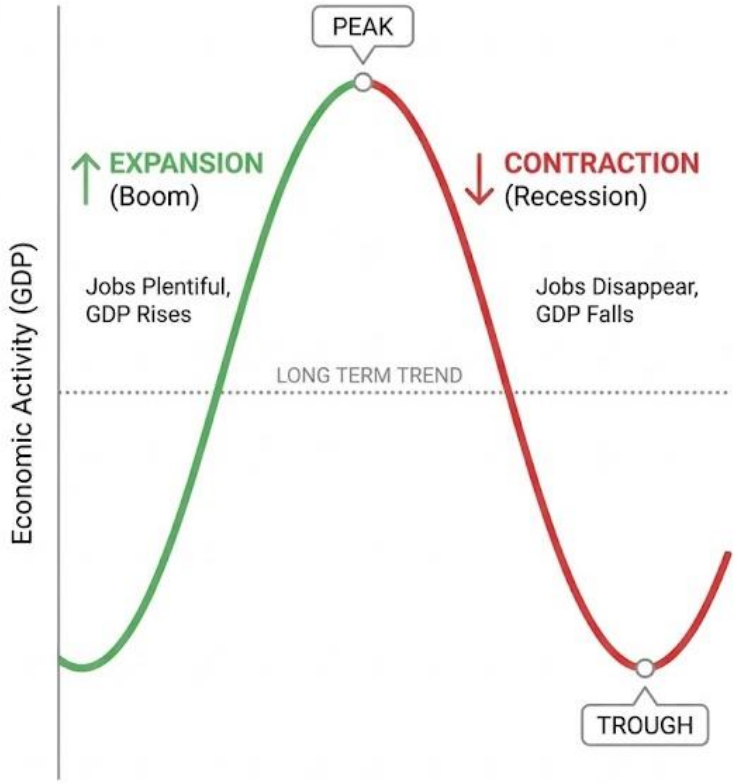
Real vs. Nominal GDP

- **Nominal GDP:** Measured in current prices. If prices double but production stays the same, Nominal GDP doubles. (Misleading).
- **Real GDP:** Adjusted for inflation (Constant prices). It measures actual *production*, not just price hikes.
- **Lesson:** Always look at Real GDP to judge economic health.



The Business Cycle

- Economies do not grow in a straight line. They oscillate.
- **Expansion (Boom):** Jobs are plentiful, GDP rises.
- **Peak:** The top of the cycle.
- **Contraction (Recession):** Jobs disappear, GDP falls (defined as 2 consecutive quarters of negative growth).
- **Trough:** The bottom of the cycle.



Computational Macro: Decomposing GDP

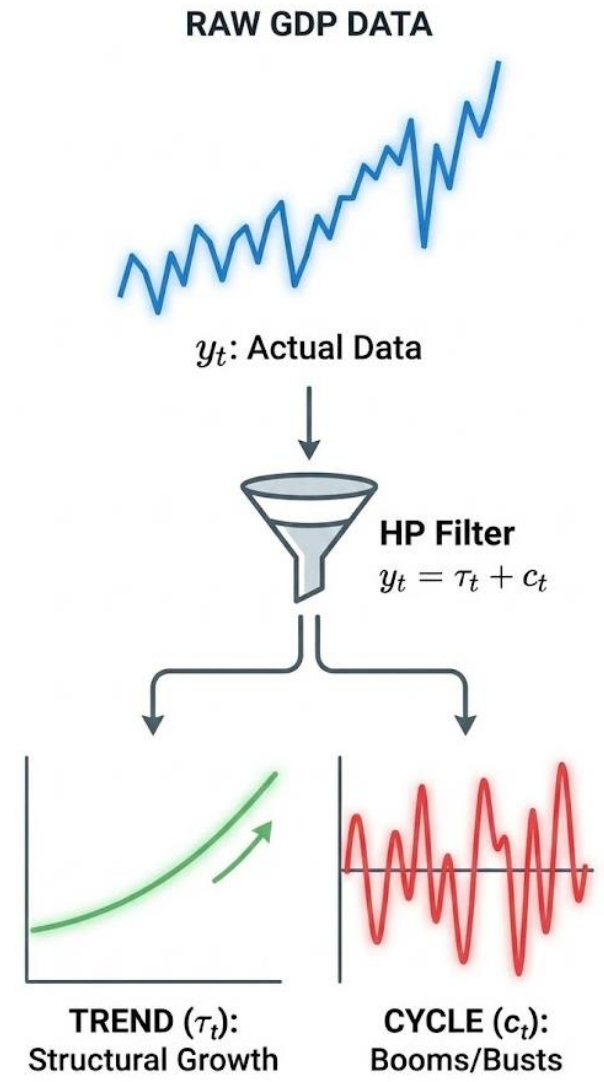
- In Data Science, we treat GDP as a **Time Series Signal**.
- **Signal Processing:** We want to separate the "Signal" (Long-term Trend) from the "Noise" (Short-term Cycle).
- **Tool:** The Hodrick-Prescott (HP) Filter.

$$y_t = \tau_t + c_t$$

y_t: Actual Data.

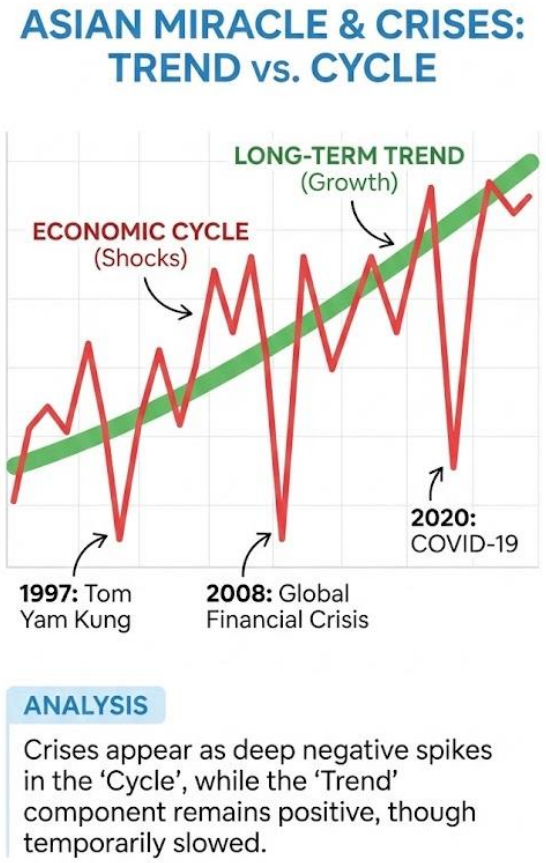
τ_t: Trend (Structural Growth).

c_t: Cycle (Booms/Busts).



Case Study: The Asian Miracle & Crises

- **The Miracle:** Rapid growth of Asian Tigers (Trend).
- **The Crises:** 1997 Tom Yam Kung, 2008 Global Financial Crisis, 2020 COVID.
- **Analysis:** These events appear as deep negative spikes in the "Cycle" component of our decomposition, while the "Trend" might remain positive (though slowed).



Part 1 Summary

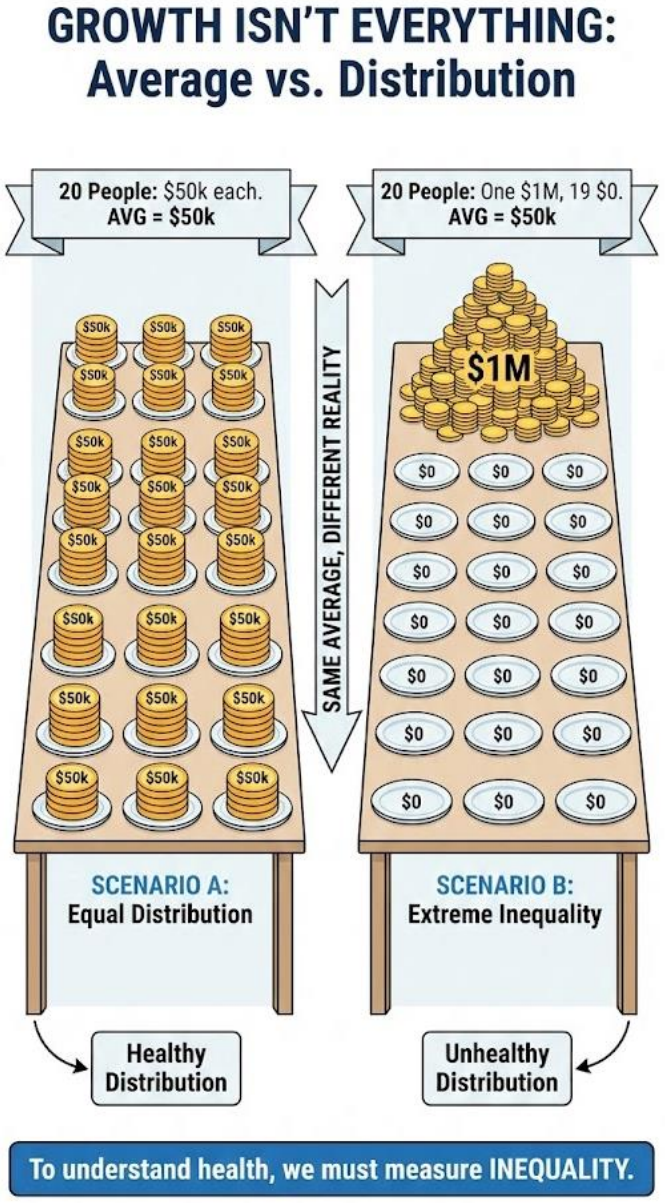
- Macroeconomics looks at the whole system.
- GDP (C+I+G+NX) is the main metric.
- Economies grow in cycles (Boom/Bust).
- We use algorithms (HP Filter) to separate short-term shocks from long-term growth.
- **Next:** How do we measure the *distribution* of this wealth?



Part 2: Inequality & Distribution

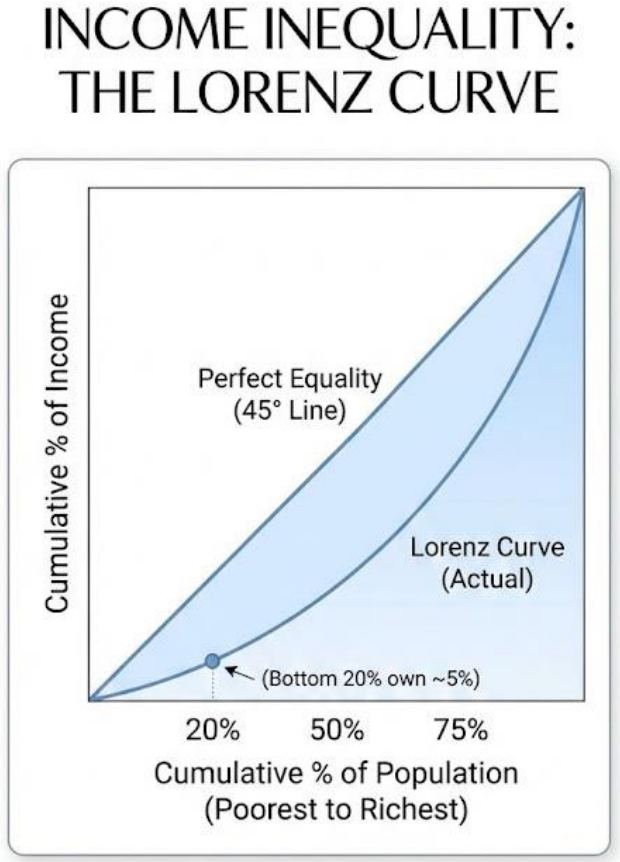
Growth isn't Everything

- **GDP per Capita:** Average income per person.
- **The Flaw:** Averages hide distribution.
  - Scenario A: Everyone earns $50k. (Avg = $50k).
  - Scenario B: One person earns $1M, 19 earn $0. (Avg = $50k).
- To understand health, we must measure **Inequality**.



The Lorenz Curve

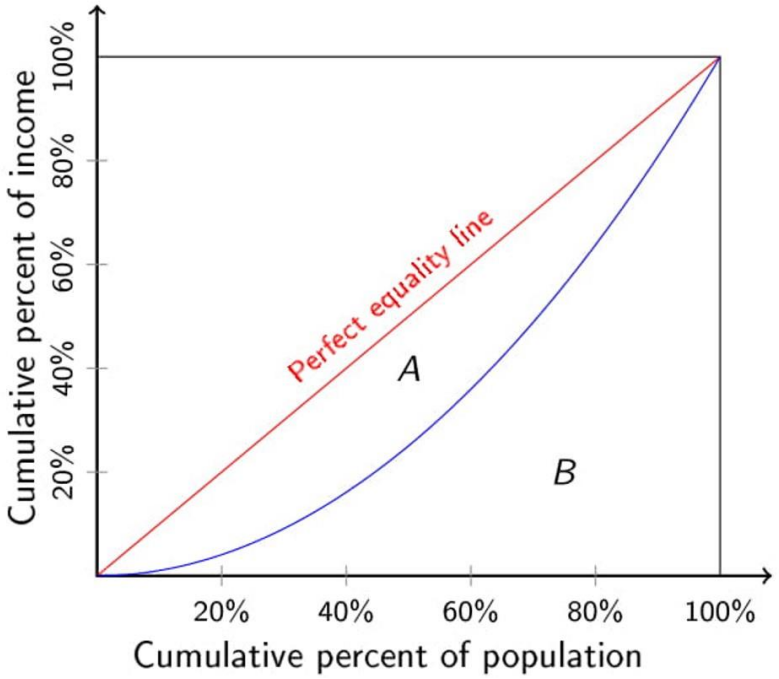
- **Definition:** A graphical representation of income distribution.
- **Axes:**
  - X-Axis: Cumulative % of Population (Poorest to Richest).
  - Y-Axis: Cumulative % of Income.
- **Line of Equality:** A 45-degree line. (Bottom 20% own 20% of income).
- **Actual Curve:** Bows underneath the equality line. (Bottom 20% might own 5%).



- Definition: Graphical representation of income distribution.

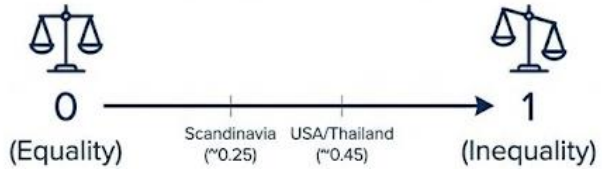
The Gini Coefficient

- **Definition:** A single number summarizing inequality based on the Lorenz Curve.
- **Calculation:** Area between Equality Line and Lorenz Curve divided by Total Area.
- **Scale:**
  - **0:** Perfect Equality (Everyone has same income).
  - **1:** Perfect Inequality (One person has everything).
- **Global Context:**
  - Scandinavia: ~0.25 (Low).
  - USA/Thailand: ~0.40-0.45 (High).
  - South Africa: ~0.63 (Highest)



https://inomics.com/terms/coeficiente-de-gini-1473340

Gini = A / (A+B)



Visualizing Inequality in Python

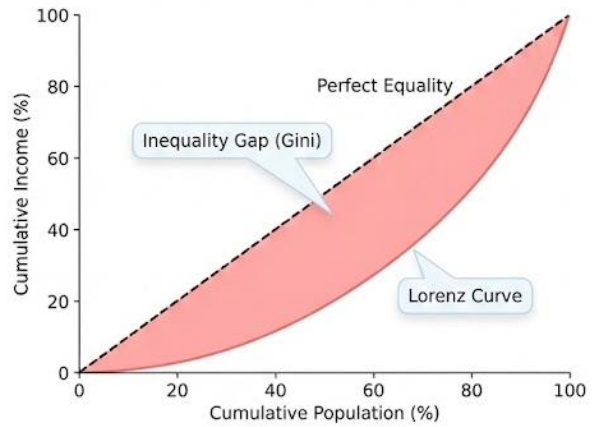
- In the tutorial, we use matplotlib to draw these curves.
- **Data Structure:** We divide the population into **Deciles** (10 groups of 10%).
- **Input:** [1, 2, 3, ... 40] (The share of income held by each 10%).
- **Output:** A visual curve and a calculated Gini score.

```
# Define Income Shares per Decile
income_shares = [1, 2, 3, 5, 7, 10, 15, 20, 25, 40]

# Calculate Cumulative Data
cumulative_pop = np.linspace(0, 100, 11)
cumulative_income = np.concatenate([0],
np.cumsum(income_shares)))

# Plot the Lorenz Curve
fig, ax = plt.subplots()
ax.plot(cumulative_pop, cumulative_income,
label='Lorenz Curve', color='#ff6b6b')
ax.plot([0, 100], [0, 100], 'k--', label='Perfect
Equality')

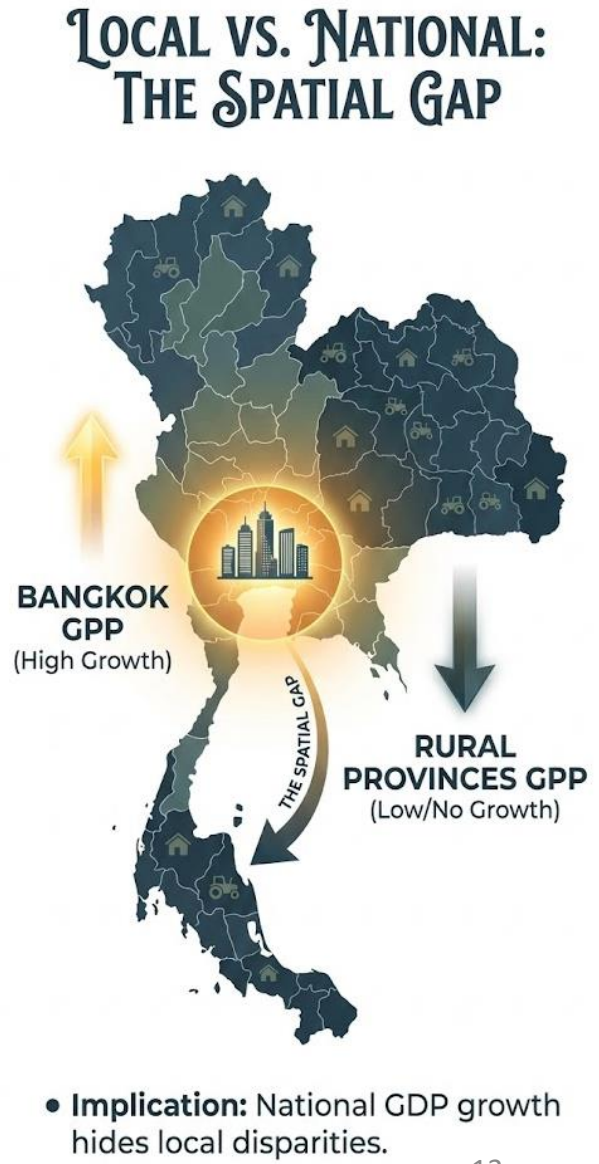
# Shade the Inequality Gap
ax.fill_between(cumulative_pop, cumulative_income,
cumulative_pop, color='#ff6b6b', alpha=0.3,
label='Inequality Gap (Gini)')
```




```
Calculated Gini Score: 0.45
```

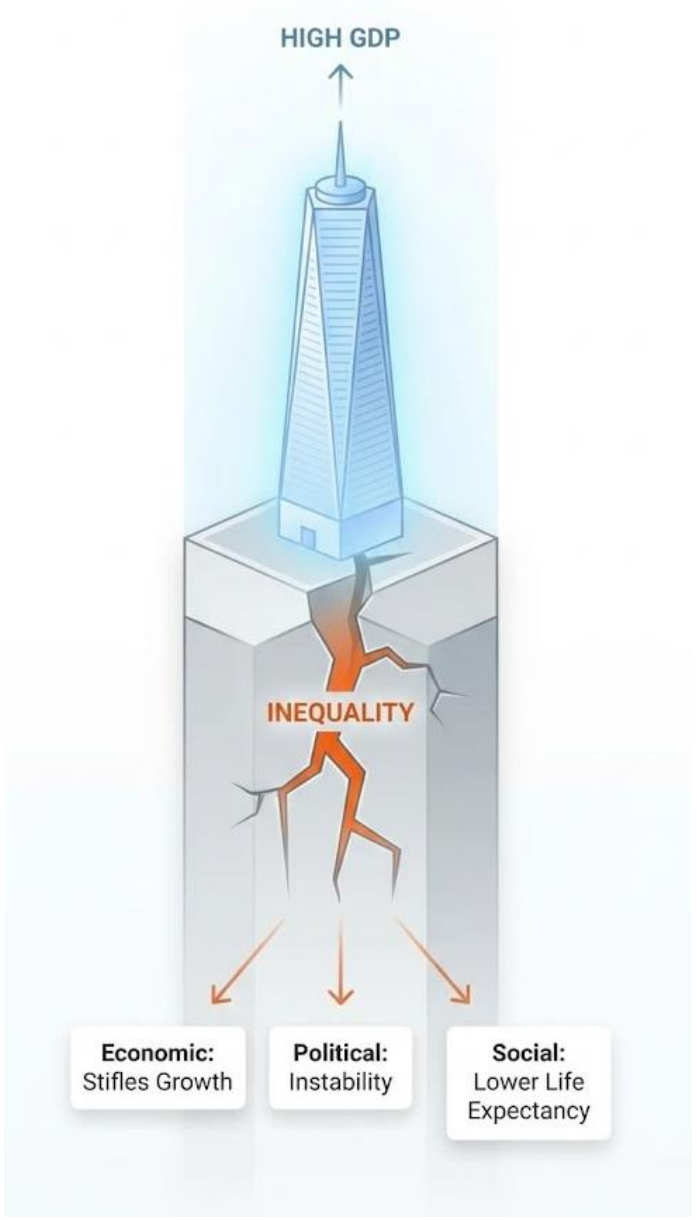
Local Economics & The Spatial Gap

- National GDP vs. Local GPP (Gross Provincial Product).
- **The "Bangkok Effect":** In many developing nations, wealth is concentrated in the capital.
- **Implication:** National GDP growth might be 5%, but rural provinces might see 0% growth.
- **Data Science Application:** We can map GPP per capita to visualize geographic inequality (Heatmaps).



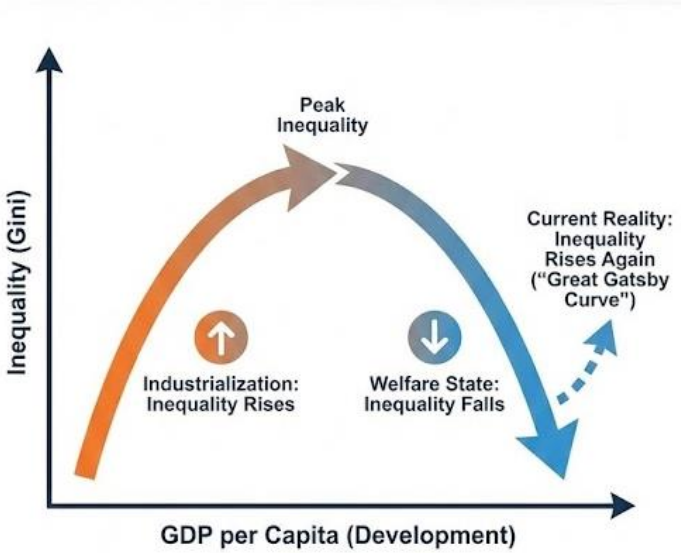
Why Inequality Matters

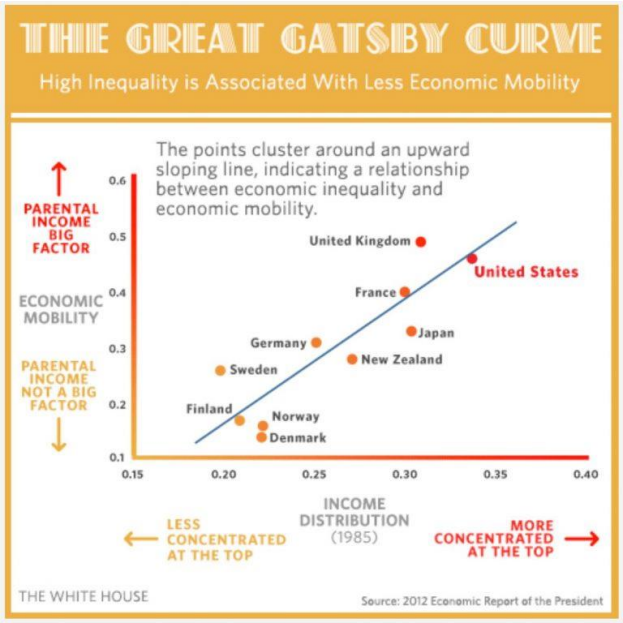
- **Economic:** High inequality can stifle growth (poor people can't invest in education/health).
- **Political:** Leads to instability and populism.
- **Social:** Correlation with crime and lower life expectancy.
- **Optimization:** Maximizing Total GDP is not enough; we must consider the *constraint* of social stability.



The Kuznets Curve Hypothesis

- **Theory:** As a country develops:
  - Inequality rises initially (Industrialization).
  - Inequality peaks.
  - Inequality falls (Service economy, welfare state).
- **Current Reality:** Many advanced economies are seeing inequality *rise* again (The "Great Gatsby Curve").
  - As inequality rises, it becomes increasingly difficult for children from poor families to move up the economic ladder compared to their parents.
  - Developing nations or those with extreme wealth gaps often appear at the top-right, where social status is most likely to be inherited.

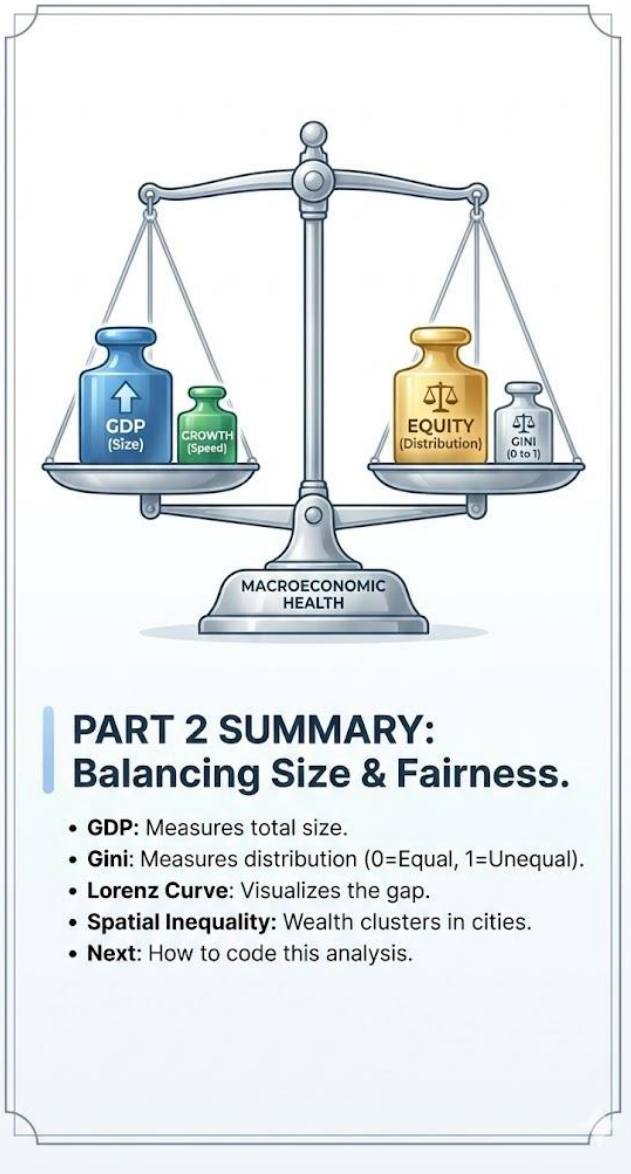




https://www.laurenzucker.org/reading-graphs-and-economic-trends-the-great-gatsby-curve-lesson/

Part 2 Summary

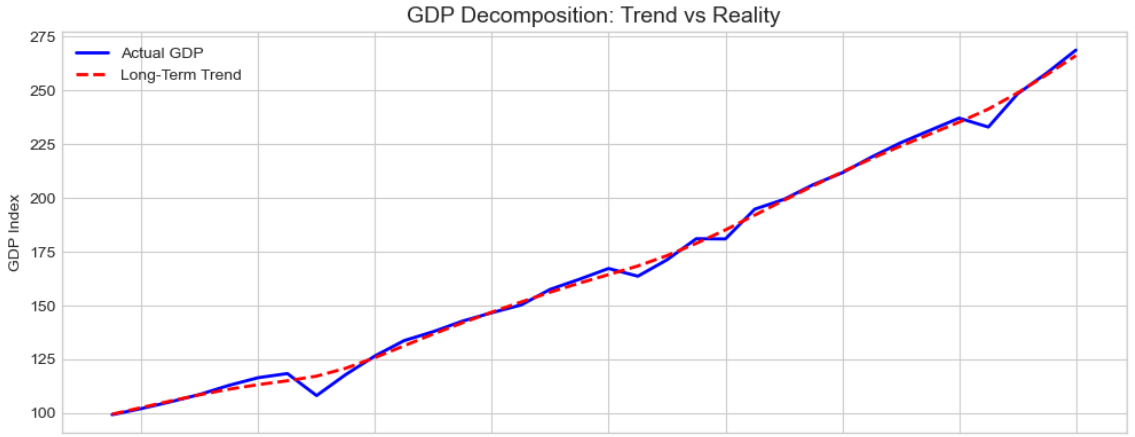
- GDP measures size; Gini measures distribution.
- **Lorenz Curve:** Visualizes the gap between the "Haves" (the rich) and "Have-Nots" (the poor).
- **Gini Coefficient:** 0 (Equal) to 1 (Unequal).
- **Spatial Inequality:** Wealth often clusters in cities.
- **Next:** How to code this analysis.

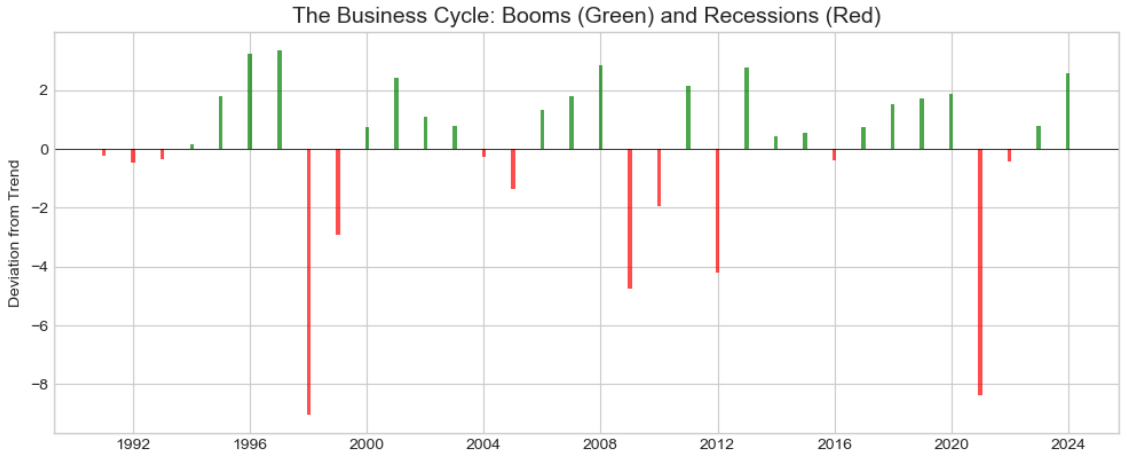


Part 3: Computational Tutorial Preview

Tutorial Overview

- **Goal:** Build a Macroeconomic Dashboard.
- **Tools:** statsmodels (for HP Filter), scipy (for Integration/Gini), matplotlib (Visualization).
- **Key Modules:**
  - **GDP Decomposition:** Extracting the Business Cycle.
  - **Inequality Analyzer:** Calculating Gini from raw data.





Generating Synthetic GDP Data

- We create a dataset that mimics Thailand's history.
- **Components:**
  - **Trend:** Exponential growth (3% per year).
  - **Shocks:** Manual adjustments for 1997 (-15%), 2008 (-8%), 2020 (-12%).
  - **Noise:** Random variance.
- **Result:** A realistic-looking jagged line graph.

GENERATING SYNTHETIC GDP DATA
– for that right on PowerPoint slide

```
# Python Code Snippet
import numpy as np

# 1. Define Components
Trend = np.exp(0.03 * time_years)  # Exponential Growth
Shocks = {1997: -0.15, 2008: -0.08, 2020: -0.12}
Noise = np.random.normal(0, 0.02, len(time_years))

# 2. Combine for Synthetic GDP
GDP_synthetic = Trend * (1 + Shocks_array) * (1 + Noise)

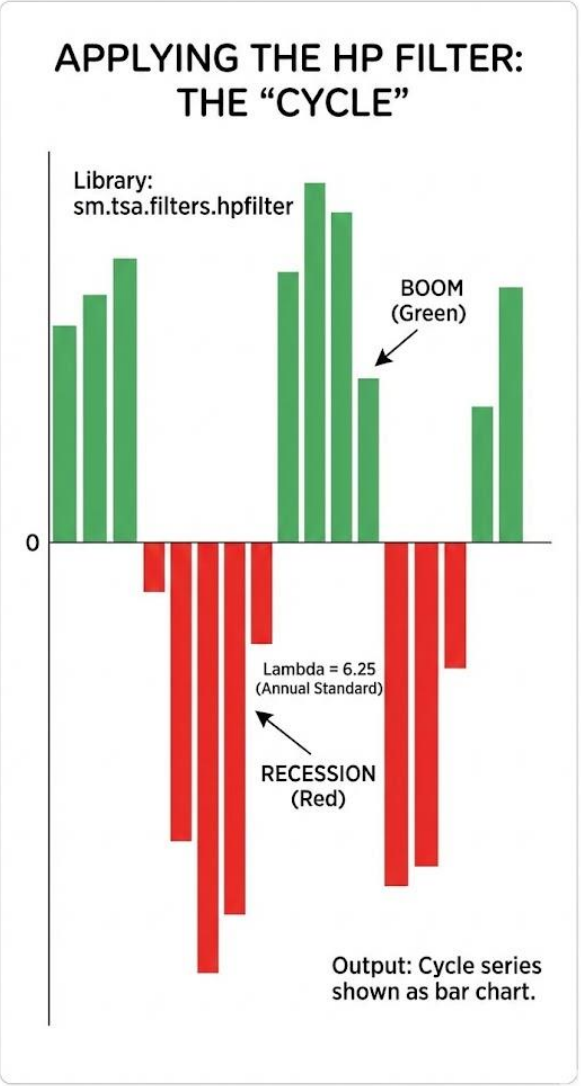
# Result: Realistic Jagged Line
```



SYNTHETIC GDP RESULT
(Realistic)

Applying the HP Filter

- **Library:** sm.tsa.filters.hpfilter
- **Parameter:** lambda=6.25 (Standard for annual data).
- **Output:** Two series - trend and cycle.
- **Visual:** We plot the Cycle as a bar chart (Red/Green) to clearly see Recessions.



Calculating Gini with Integration

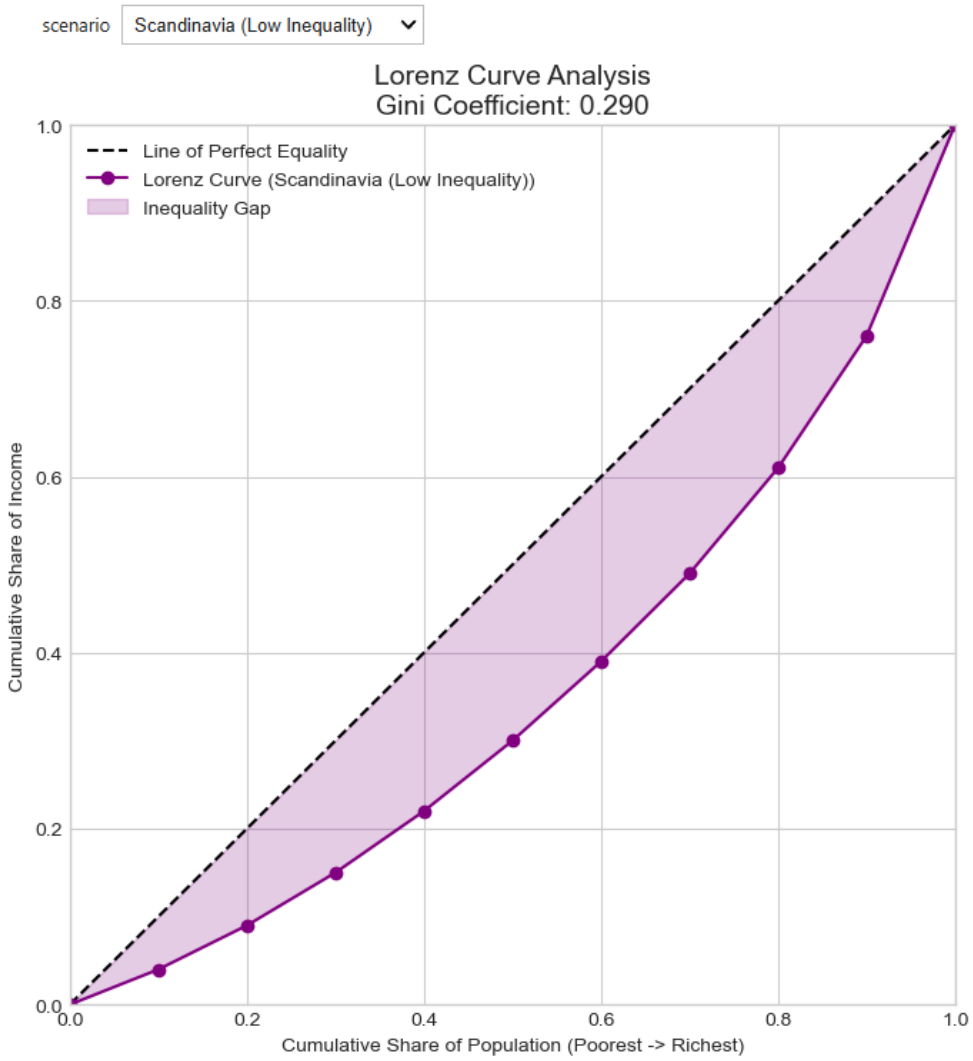
- **Math:** $\text{Gini} = 1 - 2 \times \text{Area Under Lorenz Curve.}$
- **Code:**
  scipy.integrate.trapezoid(cumulative_income, cumulative_population)

- **Logic:** We compute the area under the curve using the Trapezoidal rule, then plug it into the Gini formula.



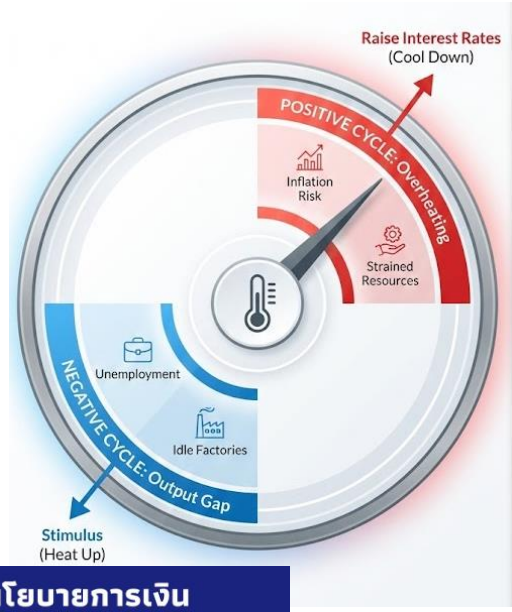
Interactive Inequality Widget

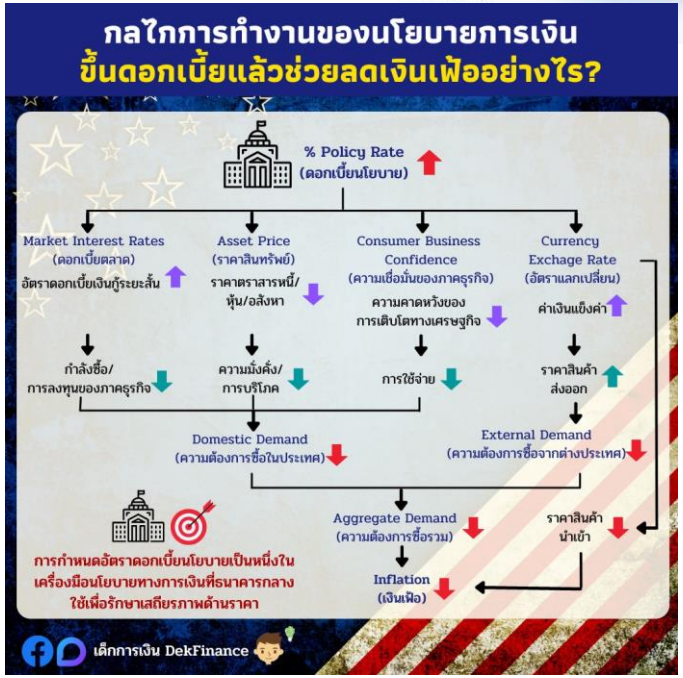
- We use ipywidgets to simulate different societies.
- **Scenarios:**
  - "Perfect Equality" (Gini = 0).
  - "Scandinavia" (Low Gini).
  - "Oligarchy" (High Gini).
- **Action:** Change the dropdown to see the Lorenz curve bow further and further away from the line of equality.



Interpreting the Cycle Component

- **Positive Cycle:** Economy is "Overheating" (Inflation risk). Resources are strained.
- **Negative Cycle:** "Output Gap" (Unemployment). Factories are idle.
- **Policy Implication:**
  - Positive → Raise Interest Rates (Cool down).
  - Negative → Stimulus (Heat up, e.g., tax cuts, interest rate cuts, and increased public spending).





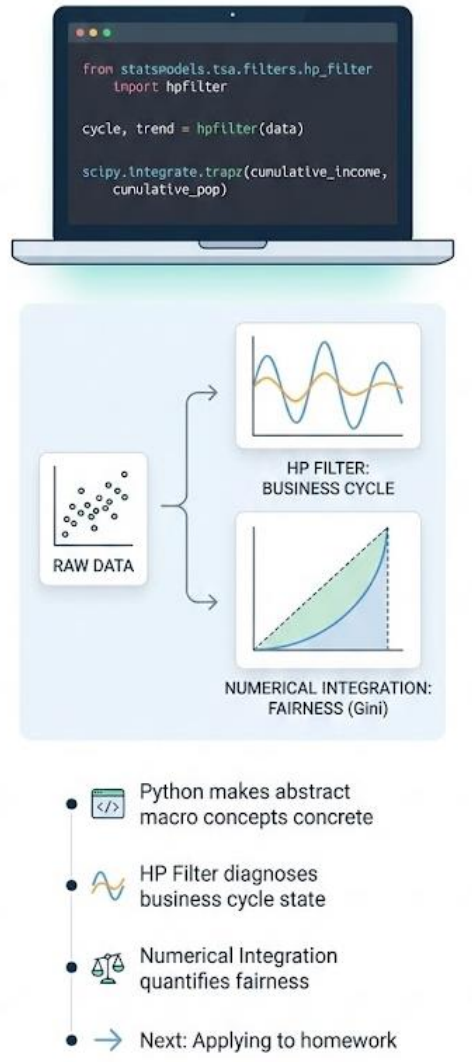
Data Sources for Real Analysis

- **GDP:** World Bank Open Data API (pandas_datareader).
- **Inequality:** World Inequality Database (WID) or National Statistical Office (NSO) Socio-Economic Surveys.
- **Challenge:** Inequality data is often sparse (collected every 2-3 years), unlike stock prices (every second).



Part 3 Summary

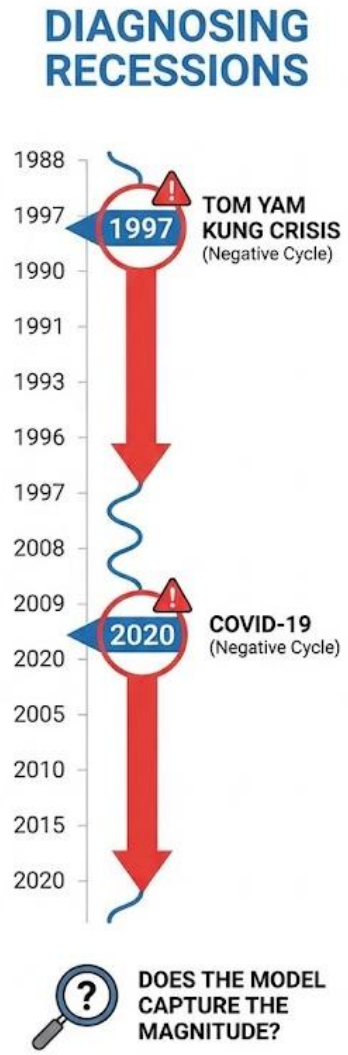
- Python makes abstract macro concepts concrete.
- HP Filter allows us to diagnose the *current* state of the business cycle.
- Numerical Integration allows us to quantify fairness.
- **Next:** Applying this to your homework.



Part 4: Homework & Conclusion

Homework Task 1: Diagnosing Recessions

- **Task:** Run the HP Filter on the provided dataset.
- **Analysis:** Look for years where the Cycle component is negative (specifically below -10).
- **Context:** Match these mathematical dips to history.
  - 1997: Tom Yam Kung Crisis.
  - 2020: COVID-19.
- **Question:** Does the model accurately capture the *magnitude* of the pain?



Hint for Task 1

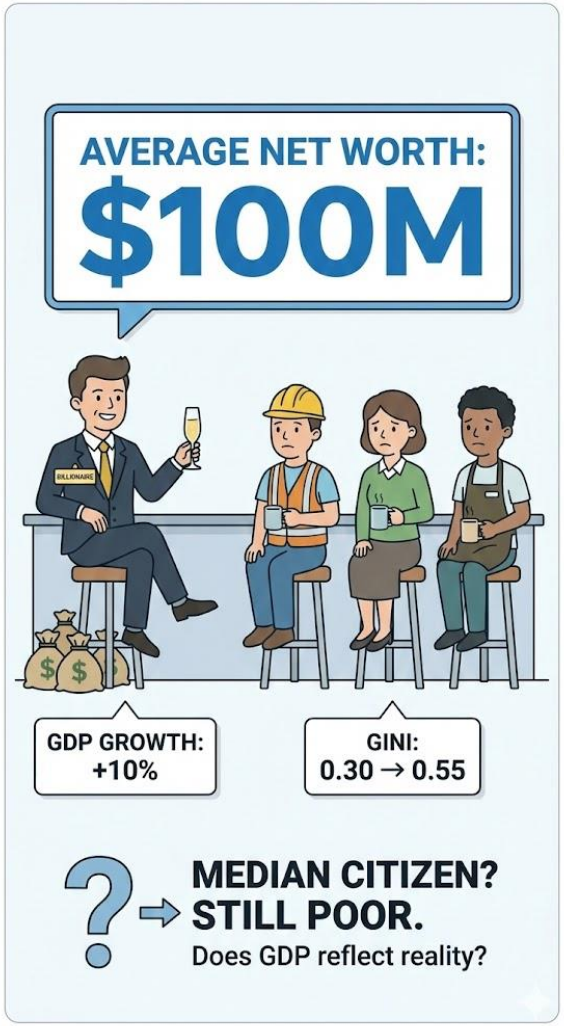
- The **Trend** line represents the "Potential GDP" (What we *could* produce if everyone worked normally).
- The **Actual** line is what *did* happen.
- A recession is the gap between Potential and Actual.



Trend = Potential GDP (Full Capacity).
Actual = What Happened.
Recession = The Gap when Actual is below Potential.

Homework Task 2: Growth vs. Inequality

- **Scenario:**
  - GDP Growth: +10% (Booming).
  - Gini: 0.30 → 0.55 (Skyrocketing inequality).
- **Task:** Analyze the impact on the **Average** vs. **Median** citizen.
- **Critical Thinking:** If Bill Gates walks into a bar, the *average* wealth skyrockets, but the *median* person is still poor. Does GDP reflect reality?



Concept: Trickle-Down vs. Inclusive Growth

- **Trickle-Down:** Focus on GDP growth; wealth will eventually reach the bottom.
  - Tax cuts and financial benefits for corporations and the wealthy will stimulate economic growth, ultimately benefiting the rest of society through job creation and increased prosperity.
- **Inclusive Growth:** Focus on broad participation; growth must reduce Gini.
  - It expands opportunities and raises living standards for all segments of the population, ensuring the benefits of prosperity are shared fairly.
- **Modern Consensus:** High inequality can hurt long-term growth (lower demand, political instability).



Final Recap

- Macroeconomics is the dashboard of the national vehicle.
- **Speedometer:** GDP Growth.
- **Check Engine:** Business Cycle (Recessions).
- **Fuel Gauge:** Inequality (Sustainability).
- Data Science gives us the tools to read these gauges accurately.



Let’s check out our Notebook

